



Covid-19 impact on political trust of European female and male citizens.

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1. EXECUTIVE SUMMARY

This study explores the impact of the COVID-19 pandemic on political trust within European political institutions, considering variables such as education, gender and age. Our hypothesis is that the level of political trust was higher in the post covid period than in the pre-covid one. In order to properly analyze the impact of the Covid-19 Pandemic on trust in political institutions in Europe the ESS round 9 – 2018 and ESS round 10 – 2020 have been selected, which can provide an insight on the situation before and after the crisis.

A comprehensive factor analysis was used to extract a single factor able to explain political trust within the countries. Moreover, the findings suggest that gender does not notably influence changes in political trust before and after Covid-19.

2. INTRODUCTION

The Covid-19 pandemic stands out as one of the most impactful and significant events in our world over the past decade, carrying consequences that will echo through future generations. In this applied research project, our goal is to investigate the dynamic fluctuations in trust within European political institutions during the Covid-19 pandemic, particularly considering potential variations across genders. Our research question is crucial as it reveals disparities in how men and women engage with political institutions during a critical exogenous event such as a pandemic, enabling targeted policy responses. Furthermore, it promotes inclusivity, equity, and a more responsive democracy, fostering informed decision-making for resilient and inclusive societies. Moreover, Social role theory provides us with a framework to help understand the complexity of gender in the political sphere. Our approach will involve conducting a comprehensive multivariate analysis of pertinent data extracted from the European Social Survey (ESS) Rounds 9 and 10, with our units of analysis being citizens of different gender. Building on existing literature, our hypothesis posits that political trust during the pandemic will exhibit the most significant decline among males. Our research strategy involves utilizing factor analysis and regression to scrutinize the relationships among variables and substantiate our findings.

3. BACKGROUND

The COVID-19 pandemic has not only posed unprecedented challenges to public health but has also stirred significant societal shifts, including changes in political trust. As the pandemic unfolded, perceptions of political entities and decision-making processes may have experienced distinctive changes among men and women. Concerns may arise if there is a significant divergence in political trust between genders, as this could indicate potential inequalities or a differential impact of the crisis.

The extensive literature on gender differences in risk aversion reveals that women are often portrayed as more cautious than men, exhibiting higher levels of risk aversion in various scenarios, including life-threatening situations. This caution manifests in professional contexts, financial decision-making, health-related behaviors, and environmental concerns. Experimental research indicates that women's higher risk aversion might be associated with their under-representation in positions of power, reflecting systematic inequalities in society. In the context of the COVID-19 pandemic, a hypothesis emerges that female respondents are more likely to support stringent government measures due to heightened risk aversion and concern for community well-being. Nuanced reactions in political trust patterns may be influenced by communication strategies, policy responses, and visibility of gender-specific issues.

The gender of political leaders also plays a significant role in shaping trust, with differences in leadership styles impacting individual responses during crises. Therefore, an examination of political trust in EU institutions by gender should encompass an exploration of the influence of gender-specific leadership dynamics.

4. DATA DESCRIPTION

This research uses data from the ESS, an academically driven cross-national survey that has been carried out across Europe since its inception in 2001. This survey involves conducting face-to-face interviews with newly chosen representative samples every two years.

To conduct our analysis, we gathered data from ESS round 9 which was conducted in 2018 and ESS round 10 which was conducted in 2020. It is fundamental to underline that to carry out an appropriate analysis we dropped the countries which do not appear in both datasets, that is: Austria, Cyprus, Denmark, Germany, Greece, Latvia, North Macedonia, Poland, Serbia, Spain, Sweden.

The ESS deploys a complex multistage probability sampling method to obtain a representative sample of individuals participating in European countries, whereby trained interviewers conduct in-person interviews with people from selected households. Within each selected cluster (through cluster analysis) households and individuals are randomly chosen. It's important to note that our examination was carried out with the consideration of weights. We introduced a novel weight termed 'overallwght,' which is the result of multiplying 'pspwght' (a weight encompassing post-stratification and design) to adjust for the country-level stratified sampling design, and 'pweight' (population size weight), addressing the over-representation of certain countries in the sample. The quality of the data, in terms of reliability, validity, accuracy and precision, is ensured by interviewer training, supervision and monitoring which is thoroughly carried out by the ESS. This ESS sampling method is designed to produce a representative and unbiased sample that can be used to study a wide range of social, political, and cultural issues around Europe.

The dependent variables we have selected to capture trust in the ESS are: Trust in country's parliament 'Trstprl', Trust in the legal system 'Trstlgl', Trust in the police 'Trstplc', Trust in politicians 'Trstplt', Trust in political parties 'Trstprt', Trust in the European Parliament 'Trstep', Trust in the United Nations 'Trstun'. All these variables have been measured through an anchored scale with a range from 0 to 10, where the value 0 designates '*No trust at all*' and value 10 stands for '*Complete trust*'.

The explanatory variables we are using in the regression analysis are two dummy variables. To tell whether males changed their trust more than females during the Pandemic, we created the dummy variable, where the value 0 refers to males and 1 to females.

To tell how the societal perception of political institutions has changed during the Covid-19 Pandemic we created the dummy Pre covid with value of 1 for ESS round 9 and value of 0 for ESS round 10.

Additional to the above-mentioned variables, the control variables for age 'agea' and years of education 'edyrs', which follow the assumption of normality, have been included in the regression. The first variable takes values in a range from 15 to 19 and corresponds to the respondent's age at the time of the survey, while the latter takes values in a range from 0 to 65.

Finally, we incorporated a latent variable, denoted as 'polltrust', to reflect the effect of European individual's trust on political institutions. This variable was generated through factor analysis utilizing the aforementioned 7 variables. The appropriateness of these variables is assessed through an initial screening, revealing fairly strong correlations among them in a correlation matrix (Exhibit 1)

5. MULTIVARIATE ANALYSIS

In the pursuit of addressing our research inquiry, it becomes incumbent to conduct a meticulous analysis of the variables at our disposal. The analysis will be focused on the several variables that capture political trust within the European social survey dataset.

Before performing any analysis, it is fundamental to clean the data at our disposal. Within the context of **data screening**, a paramount objective is to ascertain the presence of missing values and outliers.

Missing values, within the framework of statistical analysis, pertain to data points or observations that are conspicuously absent or exhibit incompleteness within a dataset. The scrutiny of missing values assumes a pivotal role in data analysis, as the presence of such gaps has the potential to significantly impact the precision and reliability of statistical findings. In this specific dataset, an inspection of the variables abovementioned revealed the existence of 4570 missing values for the variable Trstpri, 530 missing values for the variable Trstlgl, 110 missing values for the variable Trstplc, 306 missing values for the variable Trstplt, 384 missing values for the variable Trstprt, 158 missing values for the variable Trstep, 2193 missing values for the variable Trstun. In account of the relative insignificance of this proportion in relation to the total number of observations, it is judicious to assert that these missing values may be judiciously omitted, as their relevance is deemed nominal.

Once we dealt with missing values we continue the process of data cleaning analyzing outliers, characterized as anomalous data points that exhibit conspicuously elevated values in relation to their counterparts, warrant a distinct consideration. In this specific dataset, outliers were not present. Now, we will continue the analysis with the FA.

5.1. FACTOR ANALYSIS

Factor Analysis is an interdependence model technique to deal with highly correlated variables and its primary purpose is to define the underlying structure among the variables in the analysis. Within a set of variables, it identifies subsets of variables that are strongly correlated among each other but relatively weakly correlated with other variables. Each subset is used to form a new variable known as a latent variable or factor.

The indicator variables chosen for the analysis were the seven mentioned before.

Between the two methods available to perform factor analysis, the approach we used to summarize the data was through Principal Axis Factoring (PAF) otherwise known as Common Factor Analysis, which focuses on factoring only the common variance. Additionally, another reason why we decided to adopt such methods is because we did not have any prior knowledge about the dataset.

Before looking at the factors, we checked for the three assumptions (Exhibit 1) needed to perform the factor analysis. The determinant of the correlation matrix calculated is 0.005. This is satisfactory for our research as the relatively low value indicates relatively high correlations between the indicator variables exist. The absence of perfectly collinear variables which could render errors is confirmed by the non-zero determinant.

The Kaiser-Mayer-Olkin measure of sampling adequacy presented a favourable value (0.860) and the Bartlett's test for sphericity displayed a p-value equal to 0 which led us to reject the null hypothesis that the correlation matrix is identical to the identity matrix ($R=I$). Therefore, the correlation matrix is factorable, and the variables can be added to the factor analysis.

From the correlation matrix and from the heatmap (Exhibit 2), it is possible to see that these variables have a reasonably strong association. This leads us, according to the Latent Root

Criterion, to choose factors for which the eigenvalue is greater than 1, which left us with only Factor 1 (Exhibit 3b). In fact, the chosen factor ended up explaining most of the common variance of the indicator variables. We assigned the name *poltrust* to Factor1.

Determinant of the correlation matrix	
Det	0.005
Bartlett test of sphericity	
Chi-square	3.18e+05
Degrees of freedom	21
p-value	0.000
H0: variables are not intercorrelated	
Kaiser-Meyer-Olkin Measure of Sampling Adequacy	
KMO	0.860

Exhibit 1: Determinant of the correlation matrix-Bartlett test KMO

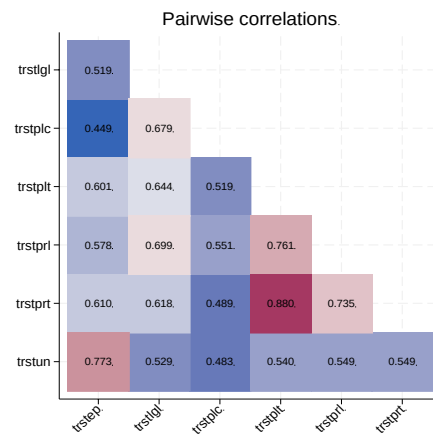


Exhibit 2: Heatplot

Variable	Factor1	Factor2	Factor3	Uniqueness
trstep	0.757	0.361	-0.132	0.278
trstlgl	0.781	-0.075	0.299	0.294
trstplc	0.662	-0.003	0.330	0.452
trstplt	0.872	-0.243	-0.184	0.147
trstpri	0.829	-0.136	0.039	0.293
trstprt	0.858	-0.211	-0.227	0.167
trstun	0.731	0.401	-0.040	0.304

Exhibit 3: Factor loadings (pattern matrix) and unique variances.

Factor	Eigenvalue	Difference	Proportion	Cumulative
Factor1	4.34181	3.92299	0.9302	0.9302
Factor2	0.41882	0.11474	0.0897	1.0200
Factor3	0.30408	0.33846	0.0651	1.0851
Factor4	-0.03439	0.04786	-0.0074	1.0778
Factor5	-0.08224	0.05265	-0.0176	1.0601
Factor6	-0.13489	0.01092	-0.0289	1.0312
Factor7	-0.14581	.	-0.0312	1.0000

Exhibit 3b: Factor analysis/correlation-principal factors unrotated

5.2. REGRESSION ANALYSIS

After constructing our political perception value through the use of factor analysis we performed a regression analysis to capture how political trust changed as a result of Covid-19. More specifically, we run a regression on the dependent variable *poltrust* and the independent dummy variable *precovid*. **Exhibit 4**, that shows a negative coefficient, leads us to conclude that the level of political trust was higher in the post-covid period than in the pre-covid one. Moreover, since the p-value is small we can conclude that our result is statistically significant.

In order to consider how the pandemic might have affected individuals in our sample according to their gender, we have created a new variable *femaleXprecovid* that shows the interaction between the gender and the timeframe of the response. **Exhibit 4** shows that there is a difference in political trust according to gender, more specifically, given the negative coefficient females show lower political trust in comparison to males in the pre-covid period. However, this relationship is not statistically significant as indicated by the p-value of 0.356 (>0.005).

poltrust	Coefficient	Std. err.	t	P> t	[95% conf.interval]	Column1
precovid	-.0362495	.0099303	-3.65	0.000	-.055713	-.016786
female	-.008111	.0098383	-0.82	0.410	-.0273941	.0111721
femaleXprecovid	-.0128874	.0139702	-0.92	0.356	-.0402691	.0144942
_cons	-.0078516	.0069987	-1.12	0.262	-.0215691	.0058659

Exhibit 4: Reg poltrust precovid female femaleXprecovid [w = overallwght]

In addition, using the power of multiple linear regression, we control for the fact that individuals in our sample differ in important ways. First, we want to see if according to their age and education, people changed their level of political trust before and after Covid-19. In order to make sure that our previous results are not meaningfully discredited, we created interactions between our *precovid* dummy and the variables *agea* (accounting for the age of the participant) and *eduyrs* (accounting for the number of full-time education years of the participant), respectively (*new_ageaXprecovid* and *new_eduyrsXprecovid*). After having run a multiple regression, we find that although higher education levels are associated with higher political trust, the pandemic has not changed the effects that education and age have on political trust, as shown (**Exhibit 5**) by the high p-values of 0.311 for *new_eduyrsXprecovid* and 0.057 for *new_ageaXprecovid*.

poltrust	Coefficient	Std. err.	t	P>t	[95% conf.interval]	Column1
new_eduyrsXprecovid	.0017834	.00176	1.01	0.311	-.0016662	.005233
new_ageaXprecovid	-.0007286	.0003834	-1.90	0.057	-.00148	.0000228
agea	-.0000329	.0002685	-0.12	0.902	-.0005593	.0004934
eduyrs	.0159277	.0012525	12.72	0.000	.0134728	.0183826
precovid	-.0237568	.0335976	-0.71	0.480	-.0896082	.0420946
_cons	-.2239687	.0239428	-9.35	0.000	-.2708966	-.1770408

Exhibit 5

6. CONCLUSION

To conclude, through a detail-oriented analysis using factor and regression analysis, the results show that our hypothesis, that the Covid epidemic positively influenced political trust levels, was confirmed. However, there is no statistically significant disparity in its impact on political trust between females and males. In other words, the gap or absence thereof in political trust among males and females has neither widened, diminished, nor become evident. Moreover, examining variables such as age and educational background in our sample has reinforced the validity of our conclusion that Covid-19 did not significantly affect the individuals in our study.

The implications of this are that European governments have been able to address both females and males effectively during the covid pandemic and were able to roll out measures that did not necessarily benefit or harm one gender over the other.

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